

Lecture 22: Free Oscillations

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Outline and ~~Motivation~~ motivation

Having obtained the equations of motion for a self-gravitating earth model, we begin our study of the Earth's **free oscillations** or **normal modes** (the terms are synonymous). The starting point is ~~developing the development of~~ a concise operator notation that closely resembles what you will know from quantum mechanics. We analyse the associated **eigenvalue problem** in a non-rotating earth model. In particular, the forced response of such an earth model is obtained as an **eigenfunction expansion**, with the frequency-domain response having **resonances** at each of the **eigenfrequencies**. It is due to this fact that ~~the~~ observations of the eigenfrequencies can be made from sufficiently long ~~time-series~~ time series, with these data providing important constraints on Earth structure. Indeed, such frequencies form the principal constraint on the Earth's density.

Linearised equations of motion

Within Lecture ~~20~~ 21 we obtained the following linearised equations of motion

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$
$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j}, \quad (1)$$

for a self-gravitating and rotating elastic earth model relative to a co-rotating reference frame. Here ρ is the equilibrium density, u_i the displacement vector, Ω_i ~~is~~ the angular velocity, A_{ijkl} the elastic tensor, and γ_i^1 the perturbation to the gravitational acceleration, which is given concretely by

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3 \mathbf{x}',$$
$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3 \mathbf{x}', \quad (2)$$

where it is understood that primed terms are functions of the integration variable, and finally $\bar{S}_{ij}$ is the stress glut associated with a seismic source. The boundary conditions for the problem are

$$\begin{aligned} \overline{A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j} &= \bar{S}_{ij} \hat{n}_j \\ \underbrace{A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j}_{\text{wavy line}} &= \bar{S}_{ij} \hat{n}_j \end{aligned} \quad (3)$$

on ∂M . While these equations express quite simple physics, they do have a complicated form, and hence it will help things greatly to find a simpler notation.

As a first step to doing this, we will express the equations in what is known as the **weak form**. Letting w_i be an arbitrary function vector field we find from eq.(1) that

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} (A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij}) - \rho \gamma_i^1 \right] w_i^* d^3 \mathbf{x} = 0.$$

(1) that

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) - \rho \gamma_i^1 \right] w_i^* d^3 \mathbf{x} = 0, \quad (4)$$

where we note that $\mathbf{w}$ has been allowed to be complex-valued and $*$ denotes a complex conjugate^{1,1}. Performing an integration by parts and using the boundary conditions this becomes

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m - \gamma_i^1 \right) w_i^* + (A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij}) \frac{\partial w_i^*}{\partial x_j} \right] d^3 \mathbf{x} = 0.$$

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m - \gamma_i^1 \right) w_i^* + \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \frac{\partial w_i^*}{\partial x_j} \right] d^3 \mathbf{x} = 0. \quad (5)$$

We have obtained this result for arbitrary $\mathbf{w}$ whenever $\mathbf{u}$ satisfies eq.(1) and (33). Conversely, the argument can be reversed to show that if eq.(5) holds for all $\mathbf{w}$, then the field $\mathbf{u}$ must satisfy eq.(1) and (33), this being the weak form of the problem.

Eq.(5) involves several terms that depend linearly on $\mathbf{u}$ and conjugate-linearly on $\mathbf{w}$, with such objects known as **sesquilinear forms**^{2,2}. For example, if we define

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \int_M \rho w_i^* u_i d^3 \mathbf{x},$$

¹ The reason for this generality is that later we will consider an associated eigenvalue problem, and in this context complex-valued fields naturally occur.

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² The prefix "sesqui" (one and a half) is used instead of "bi" to indicate that the form is linear in one argument and conjugate-linear in the other.

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \int_M \rho w_i^* u_i d^3 \mathbf{x}, \quad (6)$$

then we can write the acceleration term in the weak form as

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle, \quad (7)$$

where ~~time-derivatives~~ time derivatives are denoted using dots. ~~Hopefully the notation here should remind you of~~ The notation is intended to recall bra-ket notation. We will call $\langle \mathbf{w} | P | \mathbf{u} \rangle$ the **kinetic energy form**, and note that it may be associated with an operator, P , that acts on displacement vectors by $(P\mathbf{u})_i = \rho u_i$. In a similar manner we define

$$\begin{aligned} \langle \mathbf{w} | W | \mathbf{u} \rangle &= \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k d^3 \mathbf{x} \\ \langle \mathbf{w} | W | \mathbf{u} \rangle &= \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k d^3 \mathbf{x}, \end{aligned} \quad (8)$$

which we call the **Coriolis form**, this being associated with an operator, W , defined by $(W\mathbf{u})_i = 2\rho \epsilon_{ijk} \Omega_j u_k$. We finally have the **potential energy form**

$$\begin{aligned} \langle \mathbf{w} | H | \mathbf{u} \rangle &= \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x}, \\ \langle \mathbf{w} | H | \mathbf{u} \rangle &= \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x}, \end{aligned} \quad (9)$$

and write the force term as

$$\begin{aligned} \langle \mathbf{w} | \mathbf{f} \rangle &= \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}. \\ \langle \mathbf{w} | \mathbf{f} \rangle &= \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}. \end{aligned} \quad (10)$$

In this manner, we can concisely express the weak form of the equations of motion as

$$\begin{aligned} \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \langle \mathbf{w} | \mathbf{f} \rangle, \\ \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \langle \mathbf{w} | \mathbf{f} \rangle, \end{aligned} \quad (11)$$

which is to hold for all **test functions**, $\mathbf{w}$. From this equation it would be very tempting to write the problem in operator form as

$$P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f}.$$

$$\underline{P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f}}. \quad (12)$$

Here it is, however, important to remember that the problem's boundary conditions are incorporated within the potential energy form and force term. As a result, we should see eq. (12) as only a formal notation; the weak form incorporates the boundary conditions directly, and is generally to be preferred.

Properties of the sesquilinear forms

If we look at the kinetic energy form, then it is clear that

$$\underline{\langle \mathbf{w} | P | \mathbf{u} \rangle = \langle \mathbf{u} | P | \mathbf{w} \rangle^*},$$

$$\underline{\langle \mathbf{w} | P | \mathbf{u} \rangle = \langle \mathbf{u} | P | \mathbf{w} \rangle^*}, \quad (13)$$

and we say that the form is **self-adjoint**. Equivalently we can say that the associated operator, P , is self-adjoint. We also note that $\langle \mathbf{u} | P | \mathbf{u} \rangle$ is positive for any non-zero vector field $\mathbf{u}$, and hence the kinetic energy form defines an **inner product**. In contrast, for the Coriolis form, we find that

$$\underline{\langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, d^3 \mathbf{x} = - \left(\int_M 2\rho \epsilon_{ijk} u_i^* \Omega_j w_k \, d^3 \mathbf{x} \right)^* = - \langle \mathbf{u} | W | \mathbf{w} \rangle^*},$$

$$\underline{\langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, d^3 \mathbf{x}}$$

$$\underline{= - \left(\int_M 2\rho \epsilon_{ijk} u_i^* \Omega_j w_k \, d^3 \mathbf{x} \right)^* = - \langle \mathbf{u} | W | \mathbf{w} \rangle^*}, \quad (14)$$

this meaning that it is **anti-self-adjoint**. This implies, in particular, that $\langle \mathbf{u} | W | \mathbf{u} \rangle$ is purely imaginary for any $\mathbf{u}$.

For the potential energy form it is readily checked that the term associated with elasticity is self-adjoint due to the hyperelastic symmetry $A_{ijkl} = A_{klij}$. Similarly for the centrifugal term we have

$$\underline{\int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m \, d^3 \mathbf{x} = \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i \Omega_j \Omega_l u_m^* \, d^3 \mathbf{x} \right)^* = \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} u_i^* \Omega_j \Omega_l w_m \, d^3 \mathbf{x} \right)^*},$$

$$\underline{\int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m \, d^3 \mathbf{x} = \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} w_i \Omega_j \Omega_l u_m^* \, d^3 \mathbf{x} \right)^*}$$

$$\underline{= \left(\int_M \rho \epsilon_{ijk} \epsilon_{klm} u_i^* \Omega_j \Omega_l w_m \, d^3 \mathbf{x} \right)^*}, \quad (15)$$

where to obtain the final equality we have just relabelled and permuted the indices appropriately. What takes more work is to show that

$$\begin{aligned} \int \rho w_i^* \gamma_i^1(\mathbf{u}) d^3\mathbf{x} &= \int \rho \gamma_i^1(\mathbf{w})^* u_i d^3\mathbf{x}. \\ \int_M \rho w_i^* \gamma_i^1(\mathbf{u}) d^3\mathbf{x} &= \int_M \rho \gamma_i^1(\mathbf{w})^* u_i d^3\mathbf{x}, \end{aligned} \quad (16)$$

where we have included arguments to show which displacement vector the gravitational acceleration term corresponds to. ~~Proof~~ The proof of this identity ~~not difficult is not difficult~~, but nor is it enlightening. The end result is that the potential energy form is also self-adjoint.

Eigenfunctions and eigenvalues in a non-rotating earth model

Recall from the scaling arguments in Lecture ~~20-21~~ that while self-gravitation is important for ~~low-frequency~~ low-frequency seismology, rotation remains only a relatively small effect. For the moment let us, therefore, suppose that the earth model is non-rotating. The equations of motion then read in weak form

$$\begin{aligned} \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \langle \mathbf{w} | \mathbf{f} \rangle, \\ \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \langle \mathbf{w} | \mathbf{f} \rangle, \end{aligned} \quad (17)$$

which is to hold for any $\mathbf{w}$, and where it is understood that the centrifugal contribution to the potential energy form has been removed.

Suppose first that we set the stress glut equal to zero, and consider a time-harmonic displacement field

$$\begin{aligned} u_i(\mathbf{x}, t) &= s_i(\mathbf{x}) e^{i\omega t} \\ u_i(\mathbf{x}, t) &= s_i(\mathbf{x}) e^{i\omega t}, \end{aligned} \quad (18)$$

with $\mathbf{s}$ the spatial part of the solution, and ω an angular frequency. Putting this into eq. ~~(17)~~ (17) we arrive at

$$\begin{aligned} -\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle &= 0. \\ -\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle &= 0. \end{aligned} \quad (19)$$

This equation has the trivial solution, $\mathbf{s} = 0$, but it is interesting to ask whether it has any non-trivial solutions. As you no doubt recognise, this is an eigenvalue problem with $\mathbf{s}$ the eigenfunction and ω the eigenfrequency. Note, in particular, that the boundary conditions for the problem are built into the weak form of the equations of motion, and hence these eigenfunctions satisfy traction-free boundary conditions on ∂M .

Due to the self-adjointness of the kinetic and potential energy forms, we can quickly establish some expected properties of the eigenfunctions and eigenfrequencies. Let $\mathbf{s}_1$ be an eigenfunction associated with eigenfrequency ω_1 , and $\mathbf{s}_2$ an eigenfunction associated with eigenfrequency ω_2 . By definition we then have

$$-\omega_1^2 \langle \mathbf{w} | P | \mathbf{s}_1 \rangle + \langle \mathbf{w} | H | \mathbf{s}_1 \rangle = 0,$$

$$-\omega_1^2 \langle \mathbf{w} | P | \mathbf{s}_1 \rangle + \langle \mathbf{w} | H | \mathbf{s}_1 \rangle = 0, \quad (20)$$

for any $\mathbf{w}$. Taking this test function equal to $\mathbf{s}_2$, and using the self-adjointness of the forms, it follows that

$$0 = -\omega_1^2 \langle \mathbf{s}_2 | P | \mathbf{s}_1 \rangle + \langle \mathbf{s}_2 | H | \mathbf{s}_1 \rangle = -\omega_1^2 \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle^* + \langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle^*. \quad (21)$$

Using the eigenvalue equation for $\mathbf{s}_2$ we know that

$$\begin{aligned} \langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle &= \omega_2^2 \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle, \\ \langle \mathbf{s}_1 | H | \mathbf{s}_2 \rangle &= \omega_2^2 \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle, \end{aligned} \quad (22)$$

and hence from eq. (21) we have shown that

$$\begin{aligned} (\omega_1^2 - \omega_2^{2*}) \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle^* &= 0. \\ (\omega_1^2 - \omega_2^{2*}) \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle^* &= 0. \end{aligned} \quad (23)$$

Taking $\mathbf{s}_1 = \mathbf{s}_2$, and recalling that $\langle \mathbf{u} | P | \mathbf{u} \rangle$ is positive for any non-zero displacement field, we conclude that $\omega_1^{2*} = \omega_1^2$, which is to say that the **squared-eigenfrequencies** **squared eigenfrequencies** are real. Next, if we suppose that $\omega_1 \neq \omega_2$, then we must have

$$\begin{aligned} \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle &= 0 \\ \langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle &= 0, \end{aligned} \quad (24)$$

this meaning that eigenfunctions with different **squared-eigenfrequencies** are **squared eigenfrequencies** are orthogonal relative to the inner product defined by the kinetic energy form. Orthogonality relative to such a “**mass-weighted**” inner product should be familiar from your prior study of linear oscillations of finite-dimensional mechanical systems.

The following additional properties of these eigenfunctions and eigenfrequencies will be stated without proof^{3,3}

1. Each **squared-eigenfrequency** **squared eigenfrequency** is associated with a **finite-dimensional eigenspace**. Eigenfunctions within the same eigenspace need not be orthogonal, but we can always select a basis such that they are.

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2. There is a **countable infinity** of ~~squared-eigenfrequencies~~squared eigenfrequencies. Their values are **bounded from below**, and **accumulate only at infinity**. The latter point means that in a ~~finite-interval~~finite interval there can only be a finite number of ~~squared-eigenfrequencies~~squared eigenfrequencies.
3. The eigenfunctions form a **complete orthonormal basis** for vector fields in the earth model. This means that an arbitrary vector field can be expressed as a convergent sum of the eigenfunctions, with the necessary expansion coefficients readily determined due to their orthogonality.

To proceed, we let $k = 0, 1, \dots$ label the different squared eigenfrequencies, ω_k^2 . For each k , there is then a finite-dimensional eigenspace, and we use a second index m to label the elements of its basis. Note that the range of m can vary with k , but we leave this implicit within the notation. The m th eigenfunction belonging to the k th eigenspace will be written $|km\rangle$ in a manner familiar from quantum mechanics. These eigenfunctions can be orthonormalised relative to the kinetic energy form, this meaning that

$$\begin{aligned} \langle km|P|k'm'\rangle &= \delta_{kk'}\delta_{mm'}, \\ \langle km|P|k'm'\rangle &= \delta_{kk'}\delta_{mm'}, \end{aligned} \quad (25)$$

for all possible indices. Using this notation, we can then express the completeness of the eigenfunctions by writing an arbitrary vector field $\mathbf{u}$ ~~that satisfies the necessary boundary conditions~~ in the form

$$\begin{aligned} \mathbf{u} &= \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle, \\ \mathbf{u} &= \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle, \end{aligned} \quad (26)$$

where $\langle km|P|\mathbf{u}\rangle$ ~~are the expansion coefficients~~is the expansion coefficient of the (k, m) th term.

Eigenfunction expansions

~~A seven-day long vertical component seismogram following the a magnitude 8.2 earthquake in Bolivia in 1994. Note also that the overall amplitude decays in a roughly exponential fashion. This is due to attenuation within the Earth, but we ignore this effect within our current discussion.~~

Recall that the equations of motion in weak form are given by

$$\begin{aligned} \langle \mathbf{w}|P|\ddot{\mathbf{u}}\rangle + \langle \mathbf{w}|H|\mathbf{u}\rangle &= \langle \mathbf{w}|\mathbf{f}\rangle, \\ \langle \mathbf{w}|P|\ddot{\mathbf{u}}\rangle + \langle \mathbf{w}|H|\mathbf{u}\rangle &= \langle \mathbf{w}|\mathbf{f}\rangle, \end{aligned} \quad (27)$$

for any ~~test-function~~test function, $\mathbf{w}$. These equations can be usefully studied in the ~~frequency-domain~~frequency domain, where we use the convention

$$\tilde{\mathbf{u}}(\mathbf{x}, \omega) = \int_0^\infty \mathbf{u}(\mathbf{x}, t) e^{-i\omega t} dt, \quad \mathbf{u}(\mathbf{x}, t) = \frac{1}{2\pi} \int_{-\infty}^\infty \tilde{\mathbf{u}}(\mathbf{x}, \omega) e^{i\omega t} d\omega.$$

$$\tilde{\mathbf{u}}(\mathbf{x}, \omega) = \int_0^\infty \mathbf{u}(\mathbf{x}, t) e^{-i\omega t} dt, \quad \mathbf{u}(\mathbf{x}, t) = \frac{1}{2\pi} \int_{-\infty}^\infty \tilde{\mathbf{u}}(\mathbf{x}, \omega) e^{i\omega t} d\omega, \quad (28)$$

noting that the initial conditions that $\mathbf{u}$ and $\dot{\mathbf{u}}$ vanish ~~before time $t=0$ has for $t \leq 0$~~ have been built into the forward transformation. It is readily seen that the Fourier transformed equations of motion take the form

$$\begin{aligned} -\omega^2 \langle \mathbf{w} | P | \tilde{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \tilde{\mathbf{u}} \rangle &= \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \\ -\omega^2 \langle \mathbf{w} | P | \tilde{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \tilde{\mathbf{u}} \rangle &= \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \end{aligned} \quad (29)$$

Using the completeness of the eigenfunctions, we know that we can write

$$\begin{aligned} \tilde{\mathbf{u}} &= \sum_{k'm'} \langle k'm' | P | \tilde{\mathbf{u}} \rangle | k'm' \rangle, \\ \tilde{\mathbf{u}} &= \sum_{k'm'} \langle k'm' | P | \tilde{\mathbf{u}} \rangle | k'm' \rangle, \end{aligned} \quad (30)$$

for some frequency-dependent coefficients, $\langle k'm' | P | \tilde{\mathbf{u}} \rangle$, that are to be determined. Putting this eigenfunction expansion into eq.(29) we find

$$\begin{aligned} \sum_{k'm'} [-\omega^2 \langle \mathbf{w} | P | k'm' \rangle + \langle \mathbf{w} | H | k'm' \rangle] \langle k'm' | P | \tilde{\mathbf{u}} \rangle &= \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \\ \sum_{k'm'} [-\omega^2 \langle \mathbf{w} | P | k'm' \rangle + \langle \mathbf{w} | H | k'm' \rangle] \langle k'm' | P | \tilde{\mathbf{u}} \rangle &= \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle, \end{aligned} \quad (31)$$

which can be simplified using the eigenvalue problem satisfied by $|k'm'\rangle$ to read

$$\begin{aligned} \sum_{k'm'} (-\omega^2 + \omega_{k'}^2) \langle \mathbf{w} | P | k'm' \rangle \langle k'm' | P | \tilde{\mathbf{u}} \rangle &= \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \\ \sum_{k'm'} (-\omega^2 + \omega_{k'}^2) \langle \mathbf{w} | P | k'm' \rangle \langle k'm' | P | \tilde{\mathbf{u}} \rangle &= \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle. \end{aligned} \quad (32)$$

Finally, we take the test function equal to $|km\rangle$, and use the orthonormality condition in eq.(25) to find that the (k, m) th coefficient of $\tilde{\mathbf{u}}$ is given by

$$\begin{aligned} \langle km | P | \tilde{\mathbf{u}} \rangle &= \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} \\ \langle km | P | \tilde{\mathbf{u}} \rangle &= \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2}. \end{aligned} \quad (33)$$

In this manner, we have obtained the **eigenfunction expansion solution**

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} | km \rangle,$$

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} |km\rangle, \quad (34)$$

for the frequency-domain problem. Inversion of this solution into the time-domain can be readily done using contour integration. Such techniques are not necessary for this course, but in the final-second problem set you will be given and make use of the resulting time-domain solution.

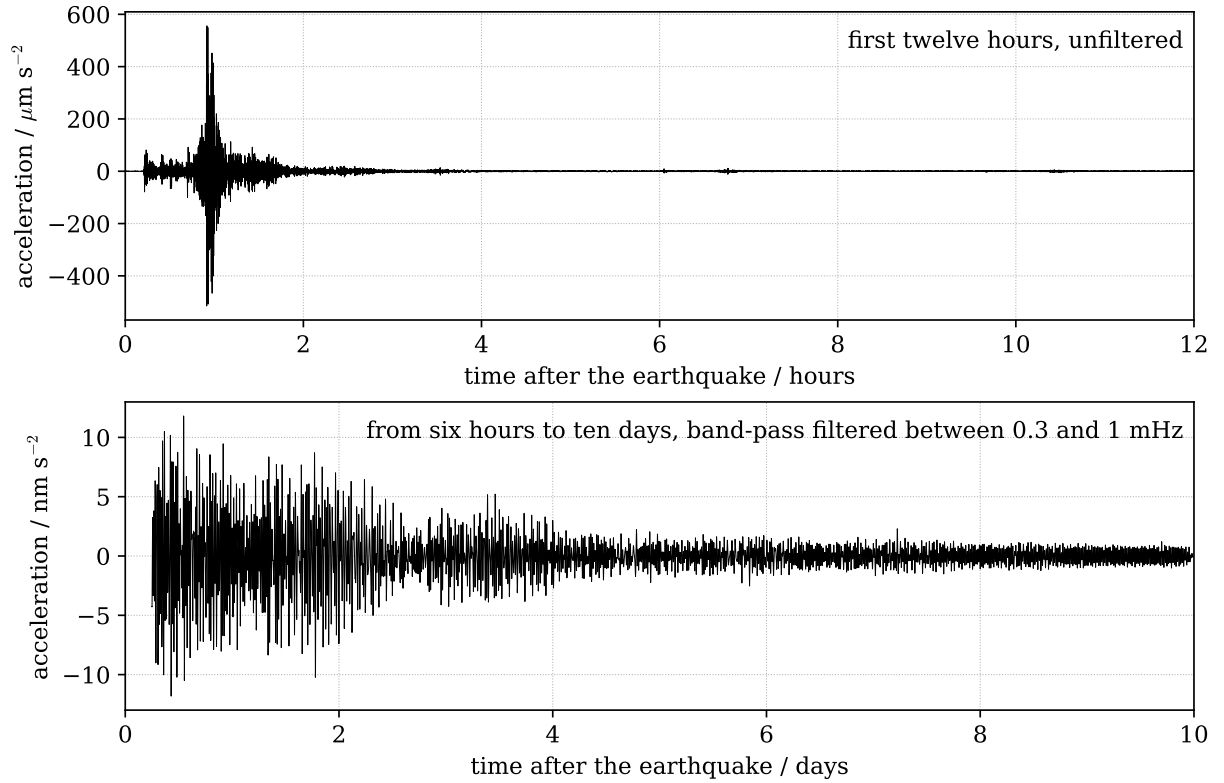


Fig. 1: Vertical-component acceleration recorded at the Black Forest Observatory in Germany following the 2011 Tohoku earthquake in Japan (moment magnitude 9.1). The amplitude spectra corresponding to Fig upper panel shows the first twelve hours of the record, dominated by the surface waves, which circle the Earth several times. 1. In some cases individual mode peaks can be identified The lower panel shows the record from six hours to ten days after the earthquake, but especially at higher-frequencies there is substantial overlap of band-pass filtered between 0.3 and 1 mHz to isolate the peaks lowest-frequency free oscillations. Their amplitude decays in a roughly exponential fashion over several days. This decay is due both to attenuation and an insufficiently long time series within the Earth, but we ignore this effect within our current discussion.

Looking at eq.(34 (34)), it is notable that there are singularities at $\pm\omega_k$ for each k . Supposing for the moment that the squared-eigenfrequencies squared eigenfrequencies are all positive, these singularities lie along the real line. Physically, we are seeing **resonances** in the frequency-domain response when the system is driven at one of its natural frequencies. From an actual seismic record we could take its Fourier transform, look for associated resonance peaks (see Fig. 2), and hence measure some of the eigen-

frequencies of the Earth. These frequencies are determined solely by Earth structure, and hence their measurement provides a possible means to avoid the **source-structure trade-off** that is ever present in seismology.

To consider such measurements further, it is important to note that recorded seismograms are not of an infinite length, and so their Fourier transform will not yield eq.(34 (34) exactly. ~~In fact, within a more complete treatment there is also slight attenuation that shifts the singularities into the upper half-plane.~~ To understand what happens with a finite-length time series, we can introduce a windowing function, $h(t)$, that is non-zero only in the range $[0, T]$, and so write the observed ~~time-series as-~~

$$\underline{\mathbf{u}_{obs}(t) = h(t)\mathbf{u}(t)}$$

time series as

$$\underline{\mathbf{u}_{obs}(t) = h(t)\mathbf{u}(t)}, \quad (35)$$

where we neglect spatial arguments for simplicity. Taking the Fourier transform of such a ~~time-series~~ time series and applying the convolution theorem we find

$$\begin{aligned} \underline{\tilde{\mathbf{u}}_{obs}(\omega) \propto \int_{-\infty}^{\infty} \tilde{\mathbf{u}}(\omega') \tilde{h}(\omega' - \omega) d\omega',} \\ \underline{\tilde{\mathbf{u}}_{obs}(\omega) \propto \int_{-\infty}^{\infty} \tilde{\mathbf{u}}(\omega') \tilde{h}(\omega - \omega') d\omega',} \end{aligned} \quad (36)$$

with the ~~right hand~~ right-hand side being proportional to the convolution of the two Fourier transforms. The end result is that the singularities of the Fourier-domain solution in eq.(34 (34) are smoothed out within real observations. The extent of the smoothing is controlled primarily by the length of the ~~time-series~~ time series, with the support of $\tilde{h}$ being largely confined to an interval whose width scales like T^{-1} . This is, of course, a familiar result; longer ~~time-series~~ time series provide better resolution in the ~~frequency-domain~~ frequency domain. Within a more complete treatment there is also slight attenuation that shifts the singularities into the upper half-plane, and this too smooths the resonance peaks.

It is for this reason that observational work on the Earth's free oscillations is concerned only with the largest earthquakes, the response to which can remain above background noise for many days, as illustrated in Fig. 1. Even so, within observed spectra we cannot always identify individual mode peaks, and this is especially true in cases of near degeneracy that we will examine in more detail in the next lecture.

Gravitational stability

Within the previous section we assumed that the squared eigenfrequencies were all positive, with the result that each term in the mode sum is associated with two resonances on the real axis, and hence oscillatory contributions to the time-domain solution. If, instead, a squared eigenfrequency, ω_k^2 , ~~was-were~~ negative then there would be two resonances on the imaginary axis, resulting in exponentially growing and decaying terms in the time-domain solution. If a squared eigenfrequency vanished, then there would be a ~~double-root~~ double root of the denominator at the origin, and this

can be shown to ~~results~~ result in a term in the time-domain solution that grows linearly with time. The underlying issue here is whether the earth model is **stable to small perturbations**, with its behaviour being **encoded within the sign of its squared eigenfrequencies**.

Recalling the form of the eigenvalue problem

$$\begin{aligned} & \underline{-\omega_k^2 \langle \mathbf{w} | P | km \rangle + \langle \mathbf{w} | H | km \rangle = 0.} \\ & \underline{-\omega_k^2 \langle \mathbf{w} | P | km \rangle + \langle \mathbf{w} | H | km \rangle = 0,} \end{aligned} \quad (37)$$

we can set the test function equal to $|km\rangle$ to obtain

$$\begin{aligned} & \underline{\omega_k^2 = \langle km | H | km \rangle.} \\ & \underline{\omega_k^2 = \langle km | H | km \rangle,} \end{aligned} \quad (38)$$

where we have made use of the eigenfunction normalisation condition. It follows that the sign of the squared eigenfrequency is set by the potential energy form.

In the non-gravitating case, this term is given by

$$\begin{aligned} & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \int_M A_{ijkl} \frac{\partial u_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} d^3 \mathbf{x},} \\ & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \int_M A_{ijkl} \frac{\partial u_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} d^3 \mathbf{x},} \end{aligned} \quad (39)$$

which is real due to the hyperelastic symmetry. We know from the first problem set that in the absence of pre-stress the elastic tensor has the additional symmetries $A_{ijkl} = A_{jikl} = A_{ijlk}$, and hence the above expression becomes

$$\begin{aligned} & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},} \\ & \underline{\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},} \end{aligned} \quad (40)$$

where we have explicitly symmetrised the spatial derivatives. We will say that the elastic tensor is pointwise stable at $\mathbf{x}$ if

$$\begin{aligned} & \underline{A_{ijkl}(\mathbf{x}) e_{ij}^* e_{kl} \geq k e_{ij}^* e_{ij}} \\ & \underline{A_{ijkl}(\mathbf{x}) e_{ij}^* e_{kl} \geq c_0 e_{ij}^* e_{ij},} \end{aligned} \quad (41)$$

for some constant $k > 0$, and all symmetric tensors e_{ij} . If such a condition holds at all points in the earth model, then we say that the elastic tensor is **uniformly pointwise stable**. Assuming this is true, we then have

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x} > 0,$$

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x} > 0, \quad (42)$$

so long as the linearised strain $e_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$ is not everywhere equal to zero. It is not difficult to show that e_{ij} vanishes identically for a non-trivial displacement vector if and only if it takes the form

$$u_i(\mathbf{x}) = a_i + A_{ij}x_j,$$

$$u_i(\mathbf{x}) = a_i + B_{ij}x_j, \quad (43)$$

with a_i a constant vector, and A_{ij} an anti-symmetric matrix. Such a displacement vector corresponds to an **infinitesimal rigid body motion**, and hence this result makes physical sense if we recall the principle of material frame indifference. It can be shown that the pointwise stability condition holds in an isotropic medium if and only if the bulk and shear moduli are both positive, and this in turn implies that P- and S-waves have positive phase speeds , with the P-wave speed being faster.

To summarise, we have seen that in the case of a non-gravitating earth model the squared-eigenfrequencies are all non-negative so long as the elastic tensor is uniformly pointwise stable. There is a six-dimensional eigenspace at zero-frequency associated with rigid body motions. Recalling, however, that the force term in the problem is

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x},$$

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}, \quad (44)$$

with $\bar{S}_{ij}$ symmetric, we see that these **trivial modes** cannot be excited by a seismic source. Thus under a reasonable restriction on the elastic tensor, we see that a non-gravitating earth model is always stable to internal perturbations.

With gravitation added to the problem there remains a six-dimensional eigenspace at zero-frequency associated with rigid body motions. The issue of stability of the remaining modes is, however, complicated by the fact that the potential energy has a gravitational contribution. While it is possible to examine this problem quantitatively, it requires some fairly advanced techniques. The physically important point, however, is that **self-gravitation is an inherently destabilising force**. Consider,

for example, a spherically symmetric and self-gravitating planet at equilibrium. If we could compress this planet uniformly, then there would be an increase in elastic potential energy. At the same time, the mass of the planet has been brought into a smaller volume, and hence the gravitational binding energy must decrease⁴.⁴ The net effect on the potential energy depends on the relative magnitude of these terms. If the elastic constants are sufficiently large, then the elastic term will dominate and the planet will be stable with respect to such a perturbation. But if gravity is sufficiently strong then the planet will keep pulling itself inwards until non-linear effects take over. A simple version of this argument for a homogeneous planet is worked through in the second problem set. In practice the question is settled numerically: for realistic earth models such as PREM, calculations of the kind shown in the next lecture return no negative squared eigenfrequencies.

What you need to know and be able to do

- (i) The weak formulation of the equations of motion using ~~bra-ket-like~~ bra-ket-like notation, and the properties of the different sesquilinear forms.
- (ii) How to derive the reality of the ~~squared-eigenfrequencies~~ squared eigenfrequencies and orthogonality of eigenfunctions in distinct eigenspaces. State without proof the remaining important properties of the eigenvalue problem.
- (iii) How to obtain a solution of the forced equations of motion in a non-rotating ~~planet~~ earth model as an eigenfunction expansion in the ~~frequency-domain~~ frequency domain. You do not need to know how to recover the time-domain solution, but may be asked to verify that such a solution works.
- (iv) That eigenfrequencies are associated with resonance peaks in the amplitude spectra obtained from seismograms, and why it is necessary to look at data from large earthquakes to obtain adequate resolution.
- (v) What it means for a non-rotating earth model to be stable in terms of the eigenfrequencies. You should understand the quantitative argument for the non-gravitating case, and the qualitative one when gravity is included.

⁴ ~~Recall that this is a negative quantity, and hence it has increased in magnitude.~~

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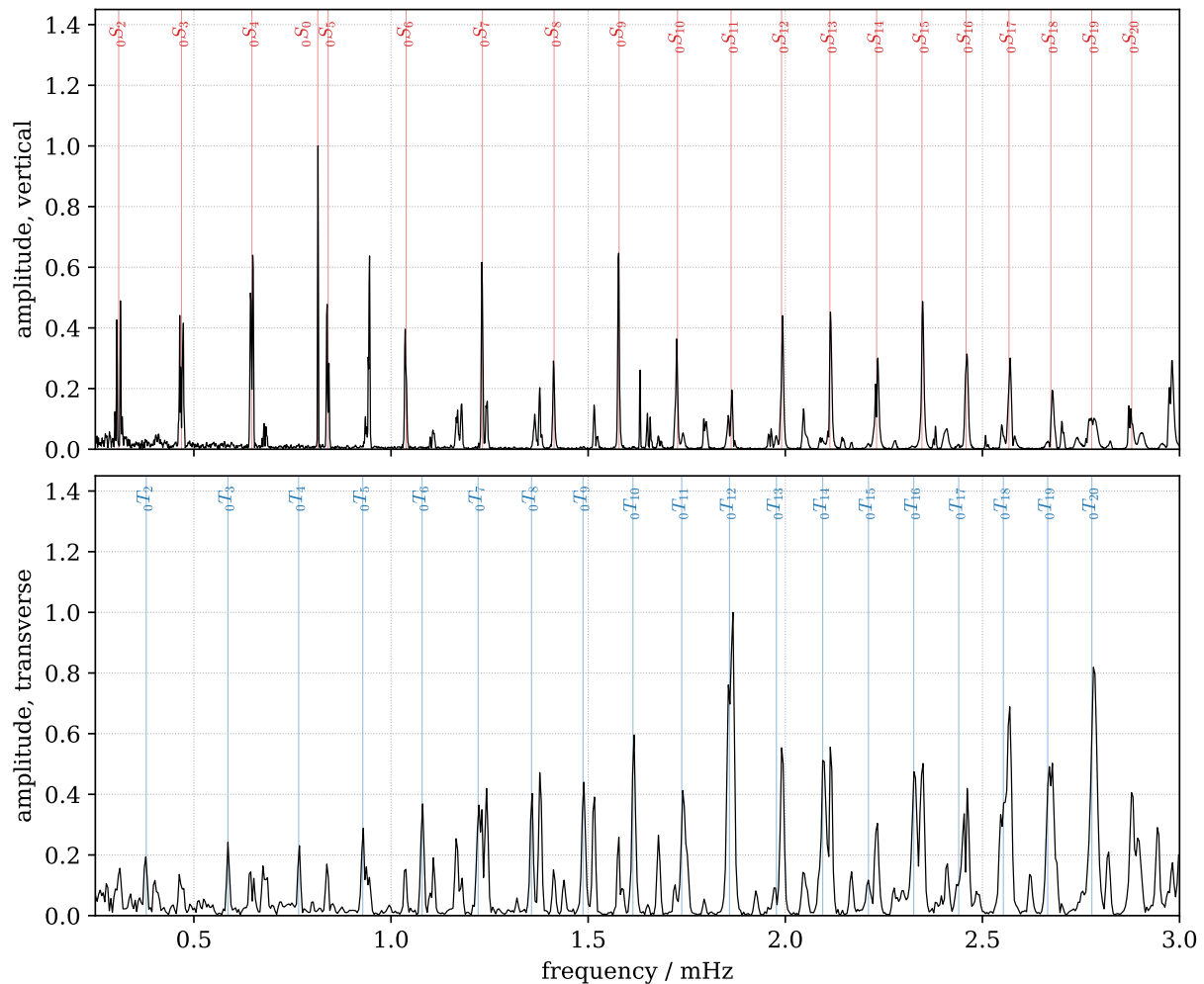


Fig. 2: Amplitude spectra of the record in Fig. 1. The upper panel is for the full ten days of the vertical component, and the lower panel for the first three days of the transverse horizontal component (perpendicular to the great circle from the earthquake), a shorter window being used because horizontal records are noisier at low frequencies. The vertical lines mark the frequencies of the fundamental spheroidal modes ${}_0S_l$ (upper panel) and toroidal modes ${}_0T_l$ (lower panel) in the earth model PREM; the largest peak in the upper panel is the radial mode ${}_0S_0$, and most of the unlabelled peaks are overtones. In some cases individual mode peaks can be identified, but at higher frequencies there is substantial overlap of the peaks due both to attenuation and an insufficiently long time series. The lowest-frequency peaks are also visibly split into several closely spaced peaks, an effect of the Earth's rotation and asphericity that we return to in the next lecture.